

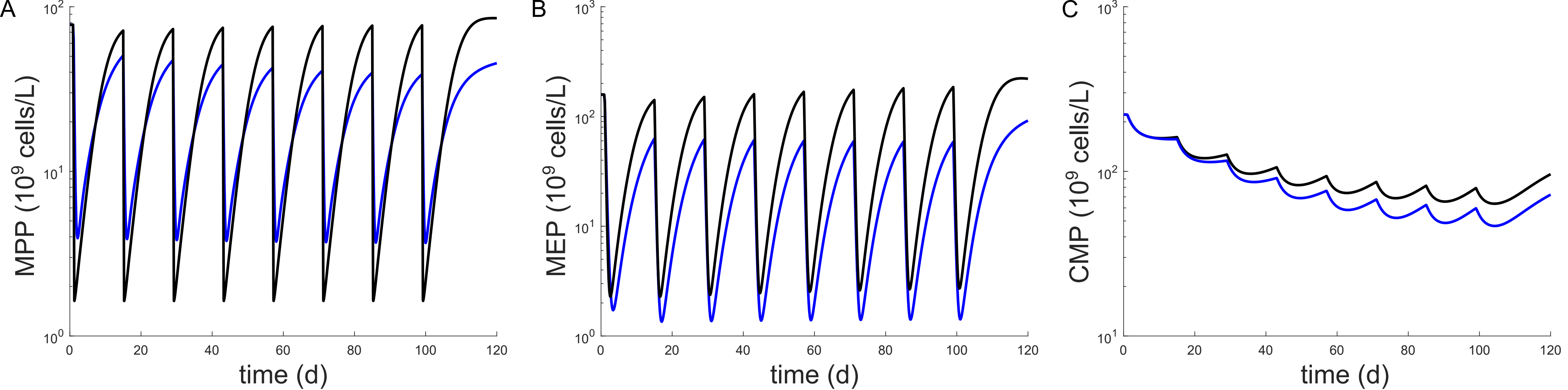


Figure S5. Alternative scenarios for competitive feedback mechanisms. Blue lines show the results of our model predictions, when we considered feedback loops from circulating cells on MPP maturation (Equation 12 main text), and feedback loops from MEP and CMP on the source feeding into MPP (Equation 11 main text). Black lines show results when we considered that feedback mechanisms regulate only the proliferation, but not the maturation, of MPP, i.e. $\text{fdbk}_{\{\text{MPP}, \text{prol}\}} = (0.5(\text{CMP}_0/\text{CMP}) + 0.5(\text{MEP}_0/\text{MEP}))^{\gamma_{\{\text{MPP}\}}}$, and also CMP and MEP proliferation, i.e. $\text{fdbk}_{\{\text{CMP}, \text{prol}\}} = (0.5(\text{Neut}_0/\text{Neut}) + 0.5(\text{Mono}_0/\text{Mono}))^{\gamma_{\{\text{CMP}\}}}$ and $\text{fdbk}_{\{\text{MEP}, \text{prol}\}} = (\text{RBC}_0/\text{RBC})^{\gamma_{\{\text{MEP}\}}}$. The dual feedback on proliferation and differentiation enabled to predict the decreasing numbers of MPP (panel A), and MEP (panel B) in successive cycles, as observed in our experiments (Figure 3 main text). Furthermore, the cell loss experienced by CMP from the 1st cycle (panel C), and their inability to recover the homeostatic equilibrium was better captured implementing the dual feedback on proliferation and differentiation into the model. Parameters used to generate black lines were estimated fitting the model with alternative feedback mechanisms only on proliferation to the data.